

Aseem Baji

Information Systems PhD Candidate | Robert H. Smith School of Business | University of Maryland
abaji@umd.edu | [GitHub](#) | [LinkedIn](#)

EDUCATION

PhD in Information Systems University of Maryland, Robert H. Smith School of Business, College Park, MD Committee: Jessica Clark (chair), Lauren Rhue, Balaji Padmanabhan, Kungpeng Zhang, Rachel Rudinger	Aug. 2022 - May 2027 (expected)
MS in Business Analytics University of Maryland, Robert H. Smith School of Business, College Park, MD GPA: 4.0/4.0; Terrapin Fellow	Sept. 2020 - Dec. 2021
BE (Hons.) in Manufacturing Birla Institute of Technology and Science Pilani, Hyderabad Campus, India GPA: 7.86/10	Aug. 2013 - July 2017

RESEARCH INTERESTS

Machine-learning pipelines; data preprocessing; information quality; predictive analytics; fairness-aware machine learning; LLM training and evaluation; responsible AI; AI for social good

DISSERTATION

Making Experimental Effort Visible in Machine-Learning Pipelines: Data Preprocessing, Fairness, and LLM Training

My dissertation examines how measuring and reporting experimental effort in pre-modeling stages can reveal interactions hidden when machine-learning pipelines are studied one component at a time. The projects develop a budgeted framework for comparing preprocessing and model search, apply this perspective to fairness, and extend it to data preparation for LLM training.

RESEARCH

JOB MARKET PAPER

Baji, A. and Clark, J. **"Measuring the Hidden Impact of Pipeline Choices: Data Preprocessing, Model Tuning, and Joint Search."** Preparing for submission to *Information Systems Research*.

WORK IN PROGRESS

Baji, A., Clark, J., and Rhue, L. **"Preprocessing Effort and Fairness in Predictive Pipelines."**

Baji, A., Wang, W., and Viswanathan, S. **"LLM-Based Personalized Math Feedback for Underserved Students."** Collaboration with the Akanksha Foundation; first-stage pilot in development, pending data access and approvals.

PROJECT IN DEVELOPMENT

Baji, A. **"Data Preprocessing and LLM Training."**

CONFERENCE PRESENTATIONS

"Measuring the Hidden Impact of Pipeline Choices: Data Preprocessing, Model Tuning, and Joint Search" INFORMS Annual Meeting, San Francisco, CA - Invited presentation	Nov. 2026 (forthcoming)
"Randomized Grid Search Experiments for Data Preprocessing" Symposium on Statistical Challenges in Electronic Commerce Research, Tokyo, Japan	June 2026
"Grid Search Experiments for Data Preprocessing" Workshop on Information Technologies and Systems, Bangkok, Thailand	Dec. 2024
"Randomized Grid Search Experiments for Data Preprocessing" Symposium on Statistical Challenges in Electronic Commerce Research, Lisbon, Portugal	June 2024
"Deep Trade-offs: Machine Learning vs. Deep Learning for Business Analytics" Symposium on Statistical Challenges in Electronic Commerce Research, Bogota, Colombia Presentations at SCECR and WITS were accepted through peer review.	June 2023

TEACHING EXPERIENCE

Instructor, BMGT 402 Database Systems

University of Maryland, Robert H. Smith School of Business

Fall 2024

Spring 2025

- Sole primary instructor for two undergraduate sections serving 70 students.
- Designed lectures, SQL labs, assignments, in-class practice, office hours, a semester-long team database-design project, and the final assessment.
- Covered SQL, SQL Server, entity-relationship modeling, relational design, normalization, query processing, data warehousing, data mining, and business information systems.

ACADEMIC SERVICE

Reviewer, International Conference on Information Systems

2026

AWARDS AND HONORS

Ritu Agarwal Endowed Fellowship - Robert H. Smith School of Business

2026

Jacob K. Goldhaber Travel Grant - University of Maryland

2026

International Conference Student Support Award - Workshop on Information Technologies and Systems

2024

Terrapin Fellow - Robert H. Smith School of Business

2020

INDUSTRY AND APPLIED EXPERIENCE

Consultant - Fractal Analytics

June 2017 - Aug. 2020

Mumbai, India

- Conducted analytics consulting for CPG and e-commerce clients using machine learning, forecasting, SQL, Tableau, and process-improvement methods.
- Developed a portfolio opportunity tool that identified approximately \$500K in growth opportunities and helped establish an Amazon Center of Excellence supporting a \$22M portfolio.
- Built forecasting, recommendation, reporting, and classification solutions; one process redesign reduced turnaround time by 25% and increased classification accuracy by 5%.

Data Science Intern / Data Scientist - University of Maryland Center for Advanced Transportation Technology

May 2021 - Aug. 2022

College Park, MD

- Applied statistical analysis and visualization to large-scale transportation and traffic-safety data.
- Developed R and Tableau analyses of traffic incidents and weather-related crashes, including a parallelized tool spanning all 52 U.S. states.

SELECTED DOCTORAL COURSEWORK

Applied Econometrics I-II | Machine Learning Applications in Business Research | Microeconomic Analysis | Machine Learning for Business Research | Applied Microeconomics | Common-Sense Reasoning and Natural Language Understanding | Structural Models | AI Design and Policy | Algorithms in Machine Learning | Responsible AI | Probabilistic Models | Information Systems in Economics | Strategy and Information Systems

TECHNICAL SKILLS

Programming: Python, R, SQL

Machine learning and analytics: Predictive modeling, classification, regression, forecasting, recommender systems, model evaluation, hyperparameter optimization, fairness analysis

Data and experimental methods: Data preprocessing, experimental design, econometrics, causal inference, A/B testing, statistical testing, reproducible computational experiments

Data systems and computing: Relational database design, SQL Server, Tableau, Power BI, high-performance computing, SLURM